COMP3670/6670
Introduction to Machine Learning
Week 5b: Vectors, Distance and Clustering

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Goals of this lecture

- Represent songs, words and images as vectors.
- Perform basic vector operations and measure distance.
- Use centroids to represent groups of datapoints.
- Understand K-Means clustering and why it terminates.
- Compare hierarchical clustering through linkage rules and dendrograms.

Vectors describe datapoints

- Computers work most comfortably with numbers, but words, songs and images have no natural numerical form.
- In ML, we choose a numerical representation that preserves information useful for the task.
- We arrange the chosen numbers as a vector.
- Linear algebra gives us tools for manipulating these vectors.

Example 1. A song can have different vector representations for different tasks.

$$\mathbf{x}_{\text{genre}} = \begin{pmatrix} 0.8 \\ 0.7 \\ 0.1 \end{pmatrix} \quad \begin{matrix} \text{danceability} \\ \text{energy} \\ \text{acoustic score} \end{matrix}$$

For engagement prediction, the same song might instead be represented by

$$\mathbf{x}_{\text{engagement}} = \begin{pmatrix} 34689 \\ 2026 \\ 721 \\ 852 \end{pmatrix} \quad \begin{matrix} \text{listens} \\ \text{likes} \\ \text{dislikes} \\ \text{comments} \end{matrix}$$

The order of the entries is part of the representation.

Example 2. Word vectors may use numerical features with no natural interpretation.

$$\text{cat} \to \begin{pmatrix} 0.3 \\ -0.8 \\ 0.9 \\ 0.1 \end{pmatrix} \quad \text{dog} \to \begin{pmatrix} 0.4 \\ -0.8 \\ 0.9 \\ -0.1 \end{pmatrix} \quad \text{bird} \to \begin{pmatrix} -0.9 \\ -0.7 \\ 1.0 \\ 0.1 \end{pmatrix} \quad \text{fly} \to \begin{pmatrix} -0.8 \\ -0.1 \\ 0.9 \\ -0.2 \end{pmatrix} \quad \text{car} \to \begin{pmatrix} 0.5 \\ 0.1 \\ -0.1 \\ 0.2 \end{pmatrix}$$

- *cat* and *dog* have similar meanings, so their vectors should be close.
- *bird* and *fly* are associated, but their meanings are not similar. Their vectors should reflect the association without being very close.
- *cat* and *car* have significantly different vectors despite their similar spellings.

Definition. A d-dimensional vector is a column of d real numbers:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix} \in \mathbb{R}^d.$$

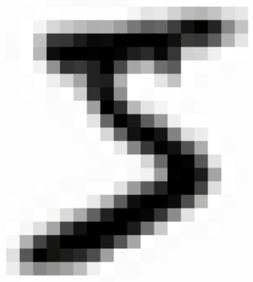
The number x_j is the j-th coordinate; d is the dimension of the representation.

A row vector is written horizontally:

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_d \end{pmatrix}.$$

Unless stated otherwise, *vector* means column vector.

Example 3. An MNIST datapoint is a 28×28 grayscale image.



List the 784 pixel intensities in a fixed order:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{784} \end{pmatrix} \in \mathbb{R}^{784}.$$

A dimension of 784 might seem high for now, but modern ML datasets often have thousands or even millions of dimensions.

Basic vector operations

Vector arithmetic. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $a \in \mathbb{R}$,

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_d + y_d \end{pmatrix}, \qquad a\mathbf{x} = \begin{pmatrix} ax_1 \\ ax_2 \\ \vdots \\ ax_d \end{pmatrix}.$$

Subtracting a vector is equivalent to adding its negation:
$\mathbf{x} - \mathbf{y} = \mathbf{x} + (-1)\mathbf{y}$.

For $a \neq 0$, scalar division is multiplication by the reciprocal:

$$\frac{\mathbf{x}}{a} = \frac{1}{a}\mathbf{x}.$$

Euclidean geometry. The Euclidean norm measures the length of a vector:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{j=1}^{d} x_j^2}.$$

The Euclidean distance between two vectors is

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2.$$

In many ML applications, a smaller distance indicates greater similarity.

Example 4. Consider

$$\mathbf{x} = \begin{pmatrix} 0.2 \\ 0.9 \\ -0.4 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} -0.3 \\ 0.8 \\ 0.5 \end{pmatrix}, \qquad \mathbf{z} = \begin{pmatrix} 0.9 \\ -0.1 \\ 0.4 \end{pmatrix}.$$

Compute $d(\mathbf{x}, \mathbf{y})$ and $d(\mathbf{y}, \mathbf{z})$. Which pair is closer?

Other norms. For $p \geq 1$, the ℓ_p norm is

$$\|\mathbf{x}\|_p = \left(\sum_{j=1}^{d} |x_j|^p \right)^{1/p}.$$

The Euclidean norm is the case $p = 2$. The limiting case is

$$\|\mathbf{x}\|_\infty = \max_{j=1,\ldots,d} |x_j|.$$

Each norm induces a distance: $d_p(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p$.

K-Means: clustering by distance

- Unsupervised learning looks for structure without labelled outcomes.
- Clustering groups datapoints that are close under the chosen representation and distance.
- Song clusters can support recommendation and exploration, while customer clusters can support segmentation.

Song dataset. 7 songs are described by three normalised features:

	danceability	energy	acoustic score
$\mathbf{x}_1$	0.91	0.08	0.37
$\mathbf{x}_2$	0.71	0.10	0.37
$\mathbf{x}_3$	0.25	0.20	0.46
$\mathbf{x}_4$	0.31	0.57	0.29
$\mathbf{x}_5$	0.11	0.59	0.34
$\mathbf{x}_6$	0.12	0.57	0.35
$\mathbf{x}_7$	0.25	0.31	0.10

Each row records the coordinates of a column vector $\mathbf{x}_i \in \mathbb{R}^3$.

Centroid. For $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^d$, their centroid is

$$\mathbf{m} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i.$$

It is the coordinate-wise average.

For songs $\mathbf{x}_1, \mathbf{x}_3, \mathbf{x}_6$ by the same singer,

$$\frac{\mathbf{x}_1 + \mathbf{x}_3 + \mathbf{x}_6}{3} = \begin{pmatrix} 0.42667 \\ 0.28333 \\ 0.39333 \end{pmatrix}.$$

The average need not be a real song, but it can represent the singer's style.

K-Means algorithm. Input: $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$ and a partition into $k \geq 2$ nonempty clusters.

1. Compute the centroid $\mathbf{c}_r$ of each cluster $\mathcal{C}_r$.
2. Assign every $\mathbf{x}_i$ to a cluster whose centroid is closest.
3. If tied, retain the current cluster.
4. Repeat until no cluster changes.

Each round alternates between updating centroid representatives and updating assignments.

Example 5. Run K-Means on the seven songs with $k = 3$ and

$$\mathcal{C}_1^{(0)} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4\},$$
$$\mathcal{C}_2^{(0)} = \{\mathbf{x}_5, \mathbf{x}_6\},$$
$$\mathcal{C}_3^{(0)} = \{\mathbf{x}_7\}.$$

	danceability	energy	acoustic score
$\mathbf{x}_1$	0.91	0.08	0.37
$\mathbf{x}_2$	0.71	0.10	0.37
$\mathbf{x}_3$	0.25	0.20	0.46
$\mathbf{x}_4$	0.31	0.57	0.29
$\mathbf{x}_5$	0.11	0.59	0.34
$\mathbf{x}_6$	0.12	0.57	0.35
$\mathbf{x}_7$	0.25	0.31	0.10

The initial centroids are

$$\mathbf{c}_1^{(0)} = \begin{pmatrix} 0.545 \\ 0.2375 \\ 0.3725 \end{pmatrix}, \quad \mathbf{c}_2^{(0)} = \begin{pmatrix} 0.115 \\ 0.58 \\ 0.345 \end{pmatrix}, \quad \mathbf{c}_3^{(0)} = \begin{pmatrix} 0.25 \\ 0.31 \\ 0.10 \end{pmatrix}.$$

Example 5 (cont'd). The first assignment gives

$$\mathcal{C}_1^{(1)} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\},$$
$$\mathcal{C}_2^{(1)} = \{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\},$$
$$\mathcal{C}_3^{(1)} = \{\mathbf{x}_7\}.$$

	danceability	energy	acoustic score
$\mathbf{x}_1$	0.91	0.08	0.37
$\mathbf{x}_2$	0.71	0.10	0.37
$\mathbf{x}_3$	0.25	0.20	0.46
$\mathbf{x}_4$	0.31	0.57	0.29
$\mathbf{x}_5$	0.11	0.59	0.34
$\mathbf{x}_6$	0.12	0.57	0.35
$\mathbf{x}_7$	0.25	0.31	0.10

The new centroids are

$$\mathbf{c}_1^{(1)} = \begin{pmatrix} 0.6233 \\ 0.1267 \\ 0.40 \end{pmatrix}, \quad \mathbf{c}_2^{(1)} = \begin{pmatrix} 0.18 \\ 0.5767 \\ 0.3267 \end{pmatrix}, \quad \mathbf{c}_3^{(1)} = \begin{pmatrix} 0.25 \\ 0.31 \\ 0.10 \end{pmatrix}.$$

Example 5 (cont'd). The second assignment gives

$$\mathcal{C}_1^{(2)} = \{\mathbf{x}_1, \mathbf{x}_2\},$$
$$\mathcal{C}_2^{(2)} = \{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\},$$
$$\mathcal{C}_3^{(2)} = \{\mathbf{x}_3, \mathbf{x}_7\}.$$

	danceability	energy	acoustic score
$\mathbf{x}_1$	0.91	0.08	0.37
$\mathbf{x}_2$	0.71	0.10	0.37
$\mathbf{x}_3$	0.25	0.20	0.46
$\mathbf{x}_4$	0.31	0.57	0.29
$\mathbf{x}_5$	0.11	0.59	0.34
$\mathbf{x}_6$	0.12	0.57	0.35
$\mathbf{x}_7$	0.25	0.31	0.10

The third-round centroids are

$$\mathbf{c}_1^{(2)} = \begin{pmatrix} 0.81 \\ 0.09 \\ 0.37 \end{pmatrix}, \quad \mathbf{c}_2^{(2)} = \begin{pmatrix} 0.18 \\ 0.5767 \\ 0.3267 \end{pmatrix}, \quad \mathbf{c}_3^{(2)} = \begin{pmatrix} 0.25 \\ 0.255 \\ 0.28 \end{pmatrix}.$$

The next assignment changes nothing, so K-Means terminates.

Analysis of K-Means

- While K-Means is quite naturally motivated, it is unclear if it must terminate.
- Each round it produces a partition of the datapoints.

  Can K-Means get stuck cycling between several partitions and never terminate?

Lemma 1. Let $\mathbf{m}$ be the centroid of $\mathbf{x}_1, \ldots, \mathbf{x}_n$. For every $\mathbf{c} \in \mathbb{R}^d$,

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \mathbf{c}\|_2^2 \;=\; \left(\sum_{i=1}^{n} \|\mathbf{x}_i - \mathbf{m}\|_2^2 \right) \;+\; n\|\mathbf{m} - \mathbf{c}\|_2^2.$$

Therefore $\mathbf{m}$ is the unique point that minimises the sum of squared distances to the datapoints.

Proof sketch.

Proof sketch (cont'd).

Theorem 2. Assume every cluster remains nonempty and ties are resolved by leaving a point in its current cluster.

Then the K-Means algorithm terminates after finitely many rounds.

K-Means objective. For nonempty clusters with centroids $\mathbf{c}_1, \ldots, \mathbf{c}_k$, the within-cluster sum of squared errors is

$$J(\mathcal{C}_1, \ldots, \mathcal{C}_k) = \sum_{r=1}^{k} \sum_{\mathbf{x}_i \in \mathcal{C}_r} \|\mathbf{x}_i - \mathbf{c}_r\|_2^2.$$

K-Means seeks a partition with small SSE.

We use SSE as a progress measure for the algorithm.

- With centroids fixed, assigning each point to a nearest centroid cannot increase the objective.
- If any point moves, the tie rule ensures that the objective decreases strictly.
- Lemma 1: With assignments fixed, replacing each centre by its cluster's centroid cannot increase the objective.
- Every round before termination strictly decreases SSE. There are only finitely many labelled partitions, so no partition can appear more than once.

- Each round computes nk distances in d dimensions, and at most k cluster averages: $\mathcal{O}(nkd)$ arithmetic operations.
- K-Means is often fast, but Vattani (2011) constructed planar instances requiring exponentially many rounds.
- An assignment can empty a cluster.

Example 6. Regard these scalars as datapoints in $\mathbb{R}^1$:

$$\mathcal{D} = \{-1.1, -1, 1, 1.1\}.$$

Initialise $\mathcal{C}_1^{(0)} = \{-1.1\}, \quad \mathcal{C}_2^{(0)} = \{1.1\}, \quad \mathcal{C}_3^{(0)} = \{-1, 1\}.$

The centroids are -1.1, 1.1 and 0.

What happens after the assignment step?

Hierarchical clustering

- K-Means requires us to choose the number of clusters k in advance.
- In many applications, we want to inspect the data before choosing the final number of groups.
- Hierarchical clustering constructs a multi-scale family of clusterings in one analysis.

Agglomerative hierarchical clustering.

1. Start with one singleton cluster per observation.
2. Find the two closest current clusters.
3. Merge them.
4. Repeat until only one cluster remains.

A linkage rule defines the distance between clusters.

Linkage rules. For nonempty clusters $\mathcal{A}$ and $\mathcal{B}$:

$$d_{\text{single}}(\mathcal{A}, \mathcal{B}) = \min_{\mathbf{x}\in\mathcal{A},\, \mathbf{y}\in\mathcal{B}} d(\mathbf{x}, \mathbf{y}),$$

$$d_{\text{complete}}(\mathcal{A}, \mathcal{B}) = \max_{\mathbf{x}\in\mathcal{A},\, \mathbf{y}\in\mathcal{B}} d(\mathbf{x}, \mathbf{y}).$$

Single linkage looks at the nearest cross-cluster pair.

Complete linkage looks at the farthest cross-cluster pair.

Linkage rules (cont'd). Average linkage uses every cross-cluster pair:

$$d_{\text{average}}(\mathcal{A}, \mathcal{B}) = \frac{1}{|\mathcal{A}|\,|\mathcal{B}|} \sum_{\mathbf{x}\in\mathcal{A}} \sum_{\mathbf{y}\in\mathcal{B}} d(\mathbf{x}, \mathbf{y}).$$

A dendrogram displays the merge sequence and merge heights.

Cutting the dendrogram at a chosen height produces a clustering.

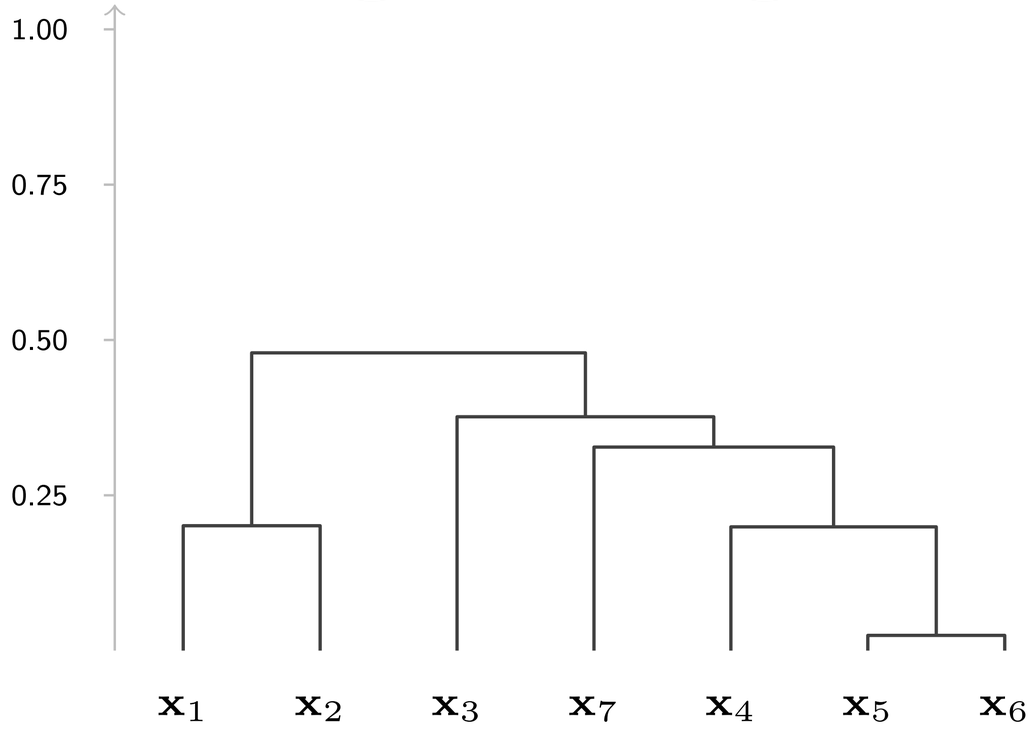
Example 7. Single linkage on the seven-song dataset begins with

	danceability	energy	acoustic score
$\mathbf{x}_1$	0.91	0.08	0.37
$\mathbf{x}_2$	0.71	0.10	0.37
$\mathbf{x}_3$	0.25	0.20	0.46
$\mathbf{x}_4$	0.31	0.57	0.29
$\mathbf{x}_5$	0.11	0.59	0.34
$\mathbf{x}_6$	0.12	0.57	0.35
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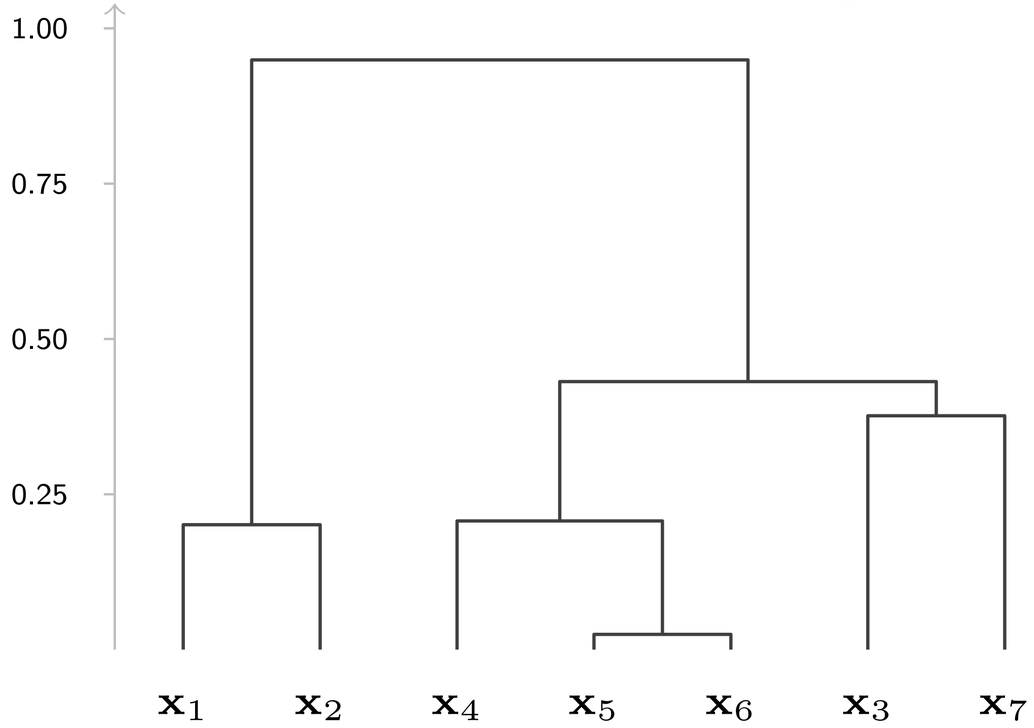
merge	height
$\mathbf{x}_5, \mathbf{x}_6$	0.0245
$\mathbf{x}_4$ joins $\{\mathbf{x}_5, \mathbf{x}_6\}$	0.1992
$\mathbf{x}_1, \mathbf{x}_2$	0.2010
$\mathbf{x}_7$ joins $\{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\}$	0.3276
$\mathbf{x}_3$ joins the right cluster	0.3764
final two clusters merge	0.4793

The dendrogram records these six merges.

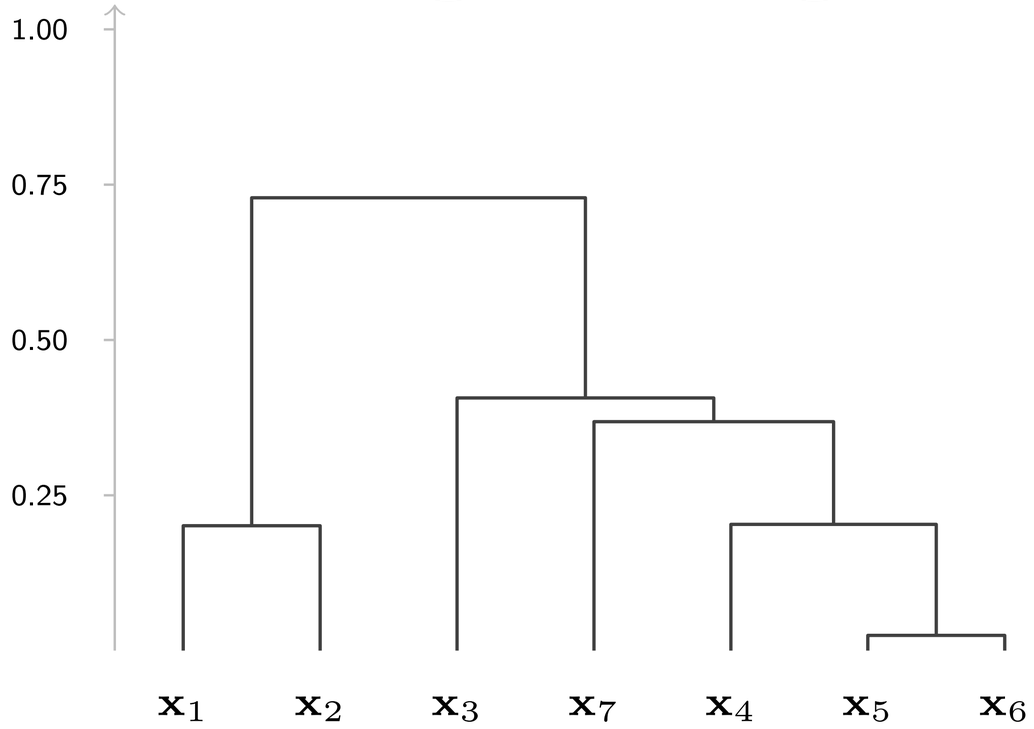
Example 7 (cont'd). Single-linkage dendrogram



Example 7 (cont'd). Complete-linkage dendrogram



Example 7 (cont'd). Average-linkage dendrogram



Example 7 (cont'd). Cutting at height 0.40 gives

linkage	clusters
single	$\{\mathbf{x}_1, \mathbf{x}_2\},\ \{\mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6, \mathbf{x}_7\}$
complete	$\{\mathbf{x}_1, \mathbf{x}_2\},\ \{\mathbf{x}_3, \mathbf{x}_7\},\ \{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6\}$
average	$\{\mathbf{x}_1, \mathbf{x}_2\},\ \{\mathbf{x}_3\},\ \{\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6, \mathbf{x}_7\}$

Single linkage can create chains. Complete linkage favours compact clusters. Average linkage provides a middle ground.

Takeaways

- Vector representations determine what algorithms can see.
- Norms and distances turn numerical differences into a notion of similarity.
- K-Means alternates centroid updates and nearest-centroid assignments for a chosen k.
- Hierarchical clustering lets us choose the number of clusters after examining a dendrogram.
- Clusters depend on the representation, distance and linkage rule; they are hypotheses, not intrinsic facts.